SUNBEL ALLOYS CO. OF INDIA LTD.
PLOT NO. D-62, TTC INDUSTRIAL AREA, TURBHE MIDC,
NAVI MUMBAI -400705, DIST. THANE. TEL. 9820096043 / 9152803284
EMAIL: roc.devashish@gmail.com , roc.enguiry@gmail.com
CIN NO. U27100MH1971PLC015004

MATERIAL SAFETY DATA SHEET

PRODUCT: TETRAHYDROFURAN

Section 1. IDENTIFICATION OF THE SUBSTANCE/PREPARATION AND OF THE COMPANY/ UNDERTAKING

1.1 Product identifier
Product name: TETRAHYDROFURAN
CAS number: 109-99-9

1.2 Relevant identified uses of the substance or mixture and uses advised against
Identified Uses:
Manufacture of Substance

1.3 Details of the supplier of the safety data sheet
Company/undertaking identification
Supplier/Manufacturer: SUNBEL ALLOYS CO. OF INDIA LTD.
Address: PLOT NO. D-62, TTC INDUSTRIAL AREA, TURBHE MIDC, NAVI MUMBAI -400705, DIST. THANE. TEL. 9820096043 / 9152803284

Section 2. HAZARDS IDENTIFICATION

Classification of the substance or mixture
Classification according to Regulation (EC) No 1272/2008
Flammable liquid, Category 2, H225
Acute toxicity, Category 4, Oral, H302
Eye irritation, Category 2, H319
Carcinogenicity, Category, H351
Specific target organ toxicity- Single Exposure, Category 3, Respiratory System, H335 For the full text of the H-Statements mentioned in this Section, see Section 16.

Signal word Danger
Hazard statement(s)
H225 Highly flammable liquid and vapour.
H302 Harmful if swallowed.
H319 Causes serious eye irritation.
H335 May cause respiratory irritation.
H351 Suspected of causing Cancer.
EUH019 May form explosive peroxides.
Precautionary statement(s)
Prevention
P210 Keep away from heat/sparks/open flames/hot surfaces. No smoking. P240 Ground/bond container and receiving equipment.

Response
P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313 IF exposed or concerned: Get medical advice / attention.
Storage
P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
Reduced labelling (≤125 ml)

Section 3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substance
Description of the substance: Tetrahydrofuran

Component	Classification	Concentration
Tetrahydrofuran		
CAS number: 109-99-9	NA	<=99%

Occupational exposure limits, if available, are listed in section 8.
See section 16 for the full text of the H-statements and R-phrases declared above.

Section 4. FIRST AID MEASURES

General advice
If symptoms persist, call a physician.
If inhaled
Fresh air. Immediately call in physician.
If Contact with skin
Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.
In case of eye contact
Rinse immediately with plenty of water, Call in ophthalmologist. Remove contact lenses.
If swallowed
Immediately make victim drink water (two glasses at most). Consult a physician.
Most important symptoms and effects, both acute and delayed:
Irritant effects, Cough, Shortness of breath, narcosis, Somnolence.

Section 5: FIRE-FIGHTING MEASURES

Extinguishing media:
Suitable Extinguishing Media- Use alcohol-resistant foam, dry chemical or carbon dioxide. Special hazards arising from the substance or mixture:
Combustible.
Forms explosive mixtures with air at ambient temperatures.
Vapours are heavier than air and may spread along floors.
Pay attention to flashback.
Development of hazardous combustion gases or vapours possible in the event of fire. Advice for firefighters:
Special protective equipment for fire-fighters
Stay in danger area only with self-contained breathing apparatus.
Prevent skin contact by keeping a safe distance or by wearing suitable protective clothing. Further Information:
Remove container from danger zone and cool with water. Prevent fire extinguishing water from contaminating surface water or the ground water system.

Section 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal protections
Advice for non-emergency personnel: Do not breathe vapours, aerosols. Avoid substance contact. Ensure adequate ventilation. Keep away from heat and sources of ignition. Evacuate the danger area, observe emergency procedures, consult an expert.
Advice for emergency responders: Protective equipment see section 8

Environmental precautions: Do not let product enter drains. Risk of explosion.

Methods and material for containment and cleaning up: If a spill or leak occurs, immediately consult your environmental supervisor. Remove ignition sources. Ventilate the area. Do not breathe the vapor or get liquid in eyes or on skin/clothing.
Dilute and flush to wastewater treatment or absorb with inert material. Do not allow the material to enter streams or waterways

Section 7: HANDLING AND STORAGE

7.1 Precautions for safe handling
Prevention of user exposure: Avice on safe handling
Use personal protective equipment. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

Conditions for safe storage, including any incompatibilities:
Stored at ambient condition, keep container tightly closed, in a well-ventilated place. Store in a cool dry area, away from heat, sparks and open flame. keep containers sealed when not in use. Store out of direct sunlight.

Incompatible materials: Organic and inorganic acid chlorides, strong oxidizing agents, alkali metals, hydrobromic acid, acidic solutions of alkali bromides.

Section 8: EXPOSURE CONTROL / PERSONAL PROTECTION

8.1 Control parameters

Control Parameters
Exposure
OSHA: No Established Limit
ACGIH: No Established Limit
NIOSH: No Established Limit
Supplier: No Established Limit
Carcinogen Data
OSHA: N/A
NTP: No known or suspended carcinogens
IARC: No known or suspended carcinogens

Exposure Controls
Respiratory: In case of mist formation use a respirator. Respirator type: organic vapor cartridge, SCBA or SAR. If respirators are used, a program should be instituted to assure compliance with OSHA standard 29 CFR 191 0.134
Eyes: Safety glasses with side shield, tight-fitting goggles or face shield.
Skin: Butyl rubber or nitrile (NBR) rubber gloves. Rubber apron and boots if splash hazard.
Engineering Controls: Provide adequate ventilation. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction. If these are not sufficient to maintain concentrations of particulates and any vapor below occupational exposure limits suitable respiratory protection must be worn.

Section 9: PHYSICAL AND CHEMICAL PROPERTIES

- Appearance: Colorless
- Physical State: Liquid
- Odor: ether-like / Petroleum distillates
- Odor Threshold: No data available
- pH: 7 to 8 at 20 °C aq solution
- Melting Point: -108 / -108.4 °C
- Critical Temperature: No data available
- Vapor Pressure: 173 hPa at 20 °C
- Relative Vapor Density: 2.5
- Auto-Ignition Temperature: 215 °C
- Minimum Ignition Energy: No data available
- Volatility: No data available
- Bulk Density: No data available

- Viscosity, dynamic: 0.47 mPa.s at 20 °C
- Water/Oil Dist. Co eff.: No data available
- Partition Co-efficient: n-octanol/Water: No data available
- Ionicity (in Water): No data available
- Lower Explosion Limit: 1.5 / 2.0 % (V)
- Upper Explosion Limit: 11.8 % (V)
- Boiling Point/Range: 66°C
- Flash Point: -21°C
- Water Solubility: at 20°C Soluble
- Molecular Weight: 72.11 g/mol

Other information:

- Molecular Formula: C4 H8 O

Section 11: TOXICOLOGICAL INFORMATION

Acute Oral toxicity LD50: 1650 mg/kg

Section 12: ECOLOGICAL INFORMATION

Toxicity to fish
Flow-through test LC50 Pimephales promelas (fathead minnow): 2,160 mg/l; 96 h
OECD Test Guideline 203
Toxicity to daphnia and other aquatic invertebrates
Static test EC50 Daphnia magna (Water flea): 3,485 mg/l; 48 h
OECD Test Guideline 202
Toxicity to algae
Cell multiplication inhibition test IC5 Scenedesmus quadricauda (Green algae): 3,700 mg/l; 8 d (maximum permissible toxic concentration)
Toxicity to bacteria
Static test IC50 activated sludge: 460 mg/l; 3 h
OECD Test Guideline 209
Toxicity to fish (Chronic toxicity)
Flow-through test NOEC Pimephales promelas (fathead minnow): 216 mg/l; 33 d
Analytical monitoring: yes
(ECHA)

Section 13: DISPOSAL CONSIDERATIONS

13.1 Waste disposal

Prevent from entering sewers or the immediate environment.
Waste must be disposed in compliance with the prevailing regulations.
Relevant disposal method: The only method authorized is collection by an authorized waste contractor and regeneration or incineration in an approved installation.

13.2 Disposal of contaminated packaging

Relevant disposal method: Empty packaging may contain flammable or explosive vapours. Hand over to an authorized contractor. Proceed in compliance with the prevailing regulations.

Section 14: TRANSPORT INFORMATION

UN number or ID number: UN 2056
UN proper shipping name: TETRAHYDROFURAN
Class: 3
Packing group: II
Environmentally hazardous: NA
Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code: NA

Section 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture
National Legislation
Storage class 3
15.2 Chemical safety assessment
For this product a chemical safety assessment was not carried out

Section 16: OTHER INFORMATION

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. The information in this document is bases on the present state of our knowledge and is applicable to the product with regard to appropriate safety precautions. It does not represent any guarantee of the properties of the product. company shall not be liable for any damage resulting from handling or from contact with the above product.